

Jainendra Bhiduri

+91 89552 34081 | johnnybhiduri8955@gmail.com | github.com/Johnnybhiduri | linkedin.com/in/jainendra-bhiduri-245054220

SUMMARY

Software Engineer II with **3 years of production experience** building AI-powered, distributed systems at scale. Designed and shipped the world's **#1 PDF accessibility platform** — used by the US Department of Justice and federal agencies — which outperforms **Adobe** in compliance scores. Expert in **Python, FastAPI, React, and AWS (Lambda, SQS/SNS, Textract, Bedrock, EC2, ElastiCache)**, with deep hands-on experience architecting **async** batch pipelines, **multi-tenant SaaS backends**, **RAG** systems, and **LLM-integrated applications**. Sole engineer on multiple government-facing products, owning full system design, implementation, and production deployment. Driven by hard engineering problems at the intersection of AI and scalable systems.

EXPERIENCE

Software Engineer II · NetraLabs · Pune, India Jan 2025 – Present
Accessibility on Demand (AOD) — US Section 508 PDF Compliance Platform · Used by the US Department of Justice & Federal Agencies

- **Rebuilt** the document-processing pipeline from synchronous to async batch execution using AWS SQS/SNS, reducing per-document processing time by **20×** (from ~60s for 3–5 pages to 25–30 pages processed in parallel) and eliminating HTTP failures under high concurrency.
- **Designed** a multi-tenant SaaS architecture (Super Admin → Reseller → Customer → User hierarchy) with JWT + 2FA authentication and per-tenant data isolation across MongoDB and S3 — the platform is now ranked #1 globally for PDF accessibility, surpassing Adobe in compliance scores.
- **Built** AI pipelines using **AWS Bedrock** and optimised LLM prompting to auto-detect and remediate complex PDF structures — tables, figures, reading order, and table of contents — that industry-standard layout detectors, including Adobe's, routinely fail to handle correctly.
- **Reduced** AI processing costs by ~40% through request batching, prompt optimisation, and selectively routing structurally simple elements away from LLM calls entirely.

Software Engineer I · NetraLabs · Pune, India Jun 2024 – Jan 2025
IMS — Secure Document Exchange System for US Correction Centers · Sole engineer

- **Delivered** a US government secure file-management system end-to-end as the sole engineer — from system design to production deployment — enabling attorney–prisoner document exchange and eliminating a physical smuggling vulnerability present in the legacy paper-based process.
- **Engineered** a React frontend with on-demand PDF loading (deferred until user click, not on page load), paginated and sortable file tables, and full Redux state management, keeping the application fast across datasets of thousands of case files.

Ground Truth — No-Code AI Agent Workflow Builder

- **Implemented** a drag-and-drop agent workflow engine (ReactFlow canvas + FastAPI) with a Decision Gate node that uses LLM classification to route conversations dynamically, and an embedded Python interpreter enabling code-execution steps inside agent pipelines.
- **Integrated RAG** pipelines using **FAISS** vector search and AWS SageMaker, allowing users to upload proprietary data, fine-tune models, and deploy AI agents within a single no-code interface.

Associate Python Developer · NetraLabs · Pune, India Jun 2023 – Jun 2024
Ground Truth platform · Remote

- **Shipped** FastAPI microservices and AWS Lambda functions handling agent orchestration, authentication, and third-party LLM API integration (OpenAI, Anthropic, Cohere) across a multi-service backend.
- **Built** the initial RAG pipeline (FAISS + OpenAI embeddings) and Redux-powered React frontend features, iterating rapidly in a fast-paced startup with weekly production deployments.

ML Intern → **"Intern of the Month" ×2** · NetraLabs (Netra AI) · Pune, India Mar 2023 – Jun 2023

- **Fine-tuned** YOLOv5 for custom object detection and deployed Flask inference APIs; built LangChain-powered chatbots for email automation, invoice parsing, and cross-platform social media posting (Facebook, Twitter, LinkedIn, Instagram).

SKILLS

Languages: Python (expert, 3 yrs), JavaScript / TypeScript, SQL

Backend: FastAPI, Flask, Django · Async Python (asyncio, TaskIQ, Celery) · Microservices · REST APIs

Frontend: React, Redux, ReactFlow, TypeScript, responsive and accessible UI

AWS: Lambda · S3 · SQS / SNS · EC2 · Textract · Bedrock · SageMaker · ElastiCache · Auto Scaling · Route 53

AI / ML: LangChain · RAG Pipelines · FAISS · GPT-4 · Claude · Fine-tuning (LLaMA, Bloom) · YOLO · HuggingFace

Data & Infra: MongoDB · Redis · Docker · GitHub Actions · Pandas · NumPy

EDUCATION

PG Diploma, Artificial Intelligence & Machine Learning · McCombs School

of Business, UT Austin, USA

Sep 2022 – Aug 2023

Supervised / Unsupervised Learning · Deep Learning · NLP · Computer Vision · Neural Networks · Recommendation Systems

B.Sc. (Hons.) Polymer Science · Delhi University — BCAS, New Delhi

Jun 2019 – Jun 2022

SELECTED IMPACT

- Platform ranked #1 globally for PDF accessibility compliance — a direct competitor to **Adobe**.
- Sole engineer on a US government project (IMS): independently designed, built, and shipped the complete system within a single product cycle.
- Diagnosed and resolved a production instability under high concurrency load; root cause was synchronous Textract polling — replaced with an event-driven SQS/SNS architecture, achieving stable operation at scale since deployment.
- Added Ratelimiter using AWS ElastiCache, to prevent request bombardment on single server.
- Added Queue management system to only allow x number of request that a system can handle.